

# Chapter 110 — Overactive Bladder

## TERMINOLOGY AND DEFINITIONS

- **Overactive bladder (OAB)** — International Continence Society (ICS): "urinary urgency, usually accompanied by frequency and nocturia, with or without urgency urinary incontinence (UUI), in the absence of urinary tract infection (UTI) or other obvious pathology."
- Component storage symptoms: **increased daytime frequency, nocturia, urgency, urgency (urinary) incontinence**. "Urge incontinence" was renamed "**urgency incontinence**" to match the abnormal sensation.

**TABLE 110.1 STANDARDIZED TERMINOLOGY AND DEFINITIONS TO DESCRIBE LOWER URINARY TRACT DYSFUNCTION RELATED TO OVERACTIVE BLADDER**

| Term                        | Definition                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------|
| OAB syndrome                | As above — <b>urgency is the defining symptom</b>                                          |
| Increased daytime frequency | Patient considers he/she voids too often by day ( <b>7 voids = upper limit of normal</b> ) |
| Nocturia                    | <b>Waking one or more times</b> to void; each void preceded and followed by sleep          |
| Urgency                     | <b>Sudden, compelling desire to void that is difficult to defer</b>                        |
| UUI                         | Involuntary loss of urine <b>associated with urgency</b>                                   |

- **Bother**: all components reduce health-related quality of life (QoL); **urgency and UUI most strongly; UUI the most bothersome LUTS in both sexes**. Frequency bothers at **≥8 daytime voids (every ≤2 h)**, though fewer can bother; **nocturia bothers at ≥2 episodes, markedly at ≥3**. **Isolated nocturia** suggests nocturnal polyuria or poor sleep, not LUT dysfunction.
- **OAB ≠ detrusor overactivity (DO)**: OAB is a **symptom syndrome**; DO is a **urodynamic observation** of involuntary detrusor contractions during filling (spontaneous or provoked). OAB patients may lack DO; DO may be asensate or an **artefact of instrumentation**.

### Mixed Symptoms Incorporating Urinary Urgency

- **Mixed urinary incontinence (MUI)** — involuntary leakage associated **with urgency and also with effort/exertion/sneezing/coughing = SUI + UUI ("OAB wet")** in one person.
- **SUI + "OAB dry" is not MUI** — describe descriptively ("SUI and urgency without UUI").
- **Urinary + faecal incontinence = double (dual) incontinence**, not MUI.
- MUI generally means **more severe leakage**; relative contributions hard to quantify, especially with **insensible incontinence** (unaware how leakage occurred).
- Fig. 110.1: incontinence spans **SUI → MUI → "OAB wet"**; "OAB dry" can coexist with SUI.

### Distinguishing Overactive Bladder From Bladder Pain Syndrome

- **Bladder pain syndrome (BPS)**: persistent/recurrent chronic **pelvic pain, pressure or discomfort perceived as bladder-related + ≥1 other urinary symptom** (urgency, frequency). Pain is **required in BPS, not typical of OAB**.

#### COMPARISON — OAB VS BPS

|                                     | OAB                                                                | BPS                                                   |
|-------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------|
| Driving fear                        | <b>Leakage</b>                                                     | <b>Pain</b>                                           |
| Sensation with filling (Fig. 110.3) | <b>Phasic surges of urgency</b> that rise and recede → void or UUI | <b>Pain rises steadily, accelerating</b> with filling |

|                  | OAB                                       | BPS                                       |
|------------------|-------------------------------------------|-------------------------------------------|
| Voided volumes   | Variable                                  | <b>More consistent</b>                    |
| Deferral         | Difficult                                 | <b>Possible, at the cost of more pain</b> |
| Site             | Perineum / base of penis / urethra–vagina | <b>Suprapubic</b> ± perineal              |
| UUI (Fig. 110.2) | Yes                                       | <b>No</b>                                 |
| Shared           | Urgency, frequency, nocturia              | Urgency, frequency, nocturia              |

- **Overlap is large** (most women with bladder-centric pain also report urgency/UUI) → probably **overlapping pathophysiology: afferent hypersensitivity, central sensitisation, somatisation.**

#### KEY POINTS: OVERACTIVE BLADDER TERMINOLOGY

- OAB = urgency ± frequency/nocturia ± UUI, without UTI or other obvious pathology.
- **OAB is symptomatic; DO is urodynamic.**
- Urgency = sudden, compelling desire to void that is difficult to defer.
- MUI = UUI + SUI.
- **"Fear of leakage" vs "fear of pain"** separates OAB from BPS, though both may express central sensitisation.

**Concept** — OAB is defined **by a sensation, urgency**, so every boundary in the terminology — DO, MUI, BPS, nocturia — is drawn by asking **what the patient feels and fears**, not what the bladder does on a trace.

## EPIDEMIOLOGY AND ECONOMICS

### Prevalence of Overactive Bladder and Related Lower Urinary Tract Symptoms

- **OAB 11.8–28.7%; UUI 1.5–17.3%** — ranges reflect differing definitions and samples.

TABLE 110.2 STUDIES EXAMINING PREVALENCE DATA FOR OVERACTIVE BLADDER

| Study                                                       | Population           | Method           | OAB definition                                                              | Prevalence                            |
|-------------------------------------------------------------|----------------------|------------------|-----------------------------------------------------------------------------|---------------------------------------|
| <b>NOBLE</b> (National Overactive Bladder Evaluation), 2003 | US, ≥18 y            | Telephone (CATI) | ≥4 urgency episodes in 4 weeks <b>and</b> ≥8 voids/day or coping behaviours | <b>OAB 16.5%; with UUI 6.1%</b>       |
| <b>EpiLUTS</b> , 2009                                       | Multinational, ≥40 y | Internet         | Urgency at least "sometimes" and/or UUI                                     | <b>OAB 28.7%; UUI 17.3%</b> (highest) |
| <b>OAB-POLL</b> , 2013                                      | US, 18–70 y          | Internet         | As EpiLUTS                                                                  | OAB 23.3%; UUI 12.6%                  |
| <b>NHANES</b> , 2008                                        | US women, ≥21 y      | Interview        | Any incontinence in 12 months                                               | UUI 7.9%; MUI 17%                     |
| <b>BACH</b> (Boston Area Community Health), 2006            | Boston, 30–79 y      | Interview        | AUA symptom index ≥8                                                        | LUTS 18.7%                            |
| <b>EPIC</b> , 2006                                          | Multinational, ≥18 y | CATI             | <b>ICS definitions</b>                                                      | <b>OAB 11.8%; UUI 1.5%</b> (lowest)   |
| Milsom, 2001                                                | Multinational, ≥40 y | Interview        | Any OAB symptom                                                             | 16.6%                                 |

CATI, computer-assisted telephone interviewing.

- **Pattern: looser definitions ("urgency at least sometimes", internet) give the highest rates; strict ICS definitions (EPIC) the lowest.** Trap: comparing raw prevalence across studies with different case definitions.
- **Sex: OAB 1.5–2× commoner in women, UUI 2–3×.** NOBLE: OAB similar by sex (~16–17%) but **UUI 7.6% women vs 2.6% men.** EpiLUTS and OAB-POLL: urgency and UUI roughly double in women.
- **Age:** rises in women from their 40s, men from their 50s–60s; UUI rises in women after ~44 and in men beyond ~64; **beyond 75 years ~40% of men and ~1/3 of women** have OAB symptoms vs ≤10% before 55. **Men overtake women in the oldest groups.**
- **Race/ethnicity:** consistently **highest in Black men and women** (EpiLUTS, OAB-POLL); NHANES — highest UUI in Black women.

### Incidence and Progression of Overactive Bladder

- **Dynamic, waxing-waning condition.** Incident OAB in women ~3–8%/year; **progression from OAB dry to UUI ~1%/year; remission ~6%/year.**

### Health Burden of Overactive Bladder

- Impairs daily activity, sleep, self-esteem, relationships, sexual life; **UUI adds burden beyond dry OAB;** less leisure physical activity.
- **Work:** higher unemployment, more absenteeism (medical and disability) and impaired productivity.
- **Care-seeking is poor:** under half ever discuss symptoms despite seeing clinicians; reasons — **not thought important, never asked, "something to cope with";** many unaware treatment exists.
- **Drug persistence is poor:** <6% take medication; antimuscarinics only **modestly superior to placebo** in aggregate; **~92% fail initial antimuscarinic within 2 years,** most stop within 6 months, half never refill. Reasons: **unmet expectations > side effects > cost.**
- **Costs:** patient costs mostly **containment (pads, laundry),** heavier with severity and in Black and Hispanic patients and those with unmet social needs; oral drug costs high, no agent universally cheap. **Third-line cost-effectiveness: sacral nerve stimulation (SNS) costliest over 10 years; onabotulinumtoxinA 100 U most cost-effective vs conservative care.** US societal cost **~\$66 billion (2007),** projected ~\$83 billion (2020).
- **Social determinants:** lower socioeconomic status associated with UUI (dose–response); food insecurity, financial strain, housing and toilet access, poor social support, interpersonal violence linked to OAB.

**Concept — OAB is common, age-related and female-predominant for UUI, yet under-reported and under-persisted — the burden lies less in lack of effective treatment than in failure to ask, expectations unmet, and cost.**

## PATHOPHYSIOLOGY AND ETIOLOGY

- **Multifactorial, across all levels of neural control.** DO was once the **sine qua non** of OAB but is no longer: **only about half with DO have urgency; 44–69% with urgency show DO.**
- **No animal model of OAB** (urgency is subjective) → research uses surrogates: **sensory activity, motor control, LUT reflexes.**
  - **Sensory:** transduction → afferent traffic → gating → sensitisation → conscious perception; **urothelium/suburothelium** release mediators, cytokines and growth factors; gut inflammation sensitises bladder nerves; brain imaging identifies candidate centres.

- **Motor:** detrusor motility and drivers (efferents, interstitial cells, mediators); **localised micromotions beside quiescent areas** summate into whole-bladder pressure.
- **Integration (Fig. 110.4):** **sacral cord** — immature reflex voiding (infants, neurologic disease); **midbrain/brainstem (PAG, PMC)** — **main regulatory level**, homeostasis, voiding reflex, switching storage ↔ voiding; **forebrain** — conscious sensation, volition, emotion, decision to store or void; **periphery** — urothelium, interstitial cells, myovesical plexus.
- **Functional brain imaging:** insula and prefrontal cortex involved; OAB alters regional activity; **SNS, tibial nerve stimulation and antimuscarinics change brain activity.**

## Afferent Mechanisms in Overactive Bladder

- Afferent endings densest **beneath the urothelium.**

### COMPARISON — BLADDER AFFERENT FIBRES

|              | Aδ fibres                                                              | C fibres                                                                                                |
|--------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Structure    | Small-calibre, myelinated                                              | Small-calibre, unmyelinated                                                                             |
| Normal role  | Sense volume and detrusor contractile state; project to higher centres | "Silent" — respond only to high-intensity stimuli (extreme distension, cold, heat, chemical irritation) |
| In pathology | —                                                                      | Become spontaneously active and hypersensitive to low-intensity input (via second-order spinal neurons) |

- **Two broad categories:** ↑ afferent signalling from the bladder, or ↓ central capacity to modulate afferent signals.

## Mechanisms of Increased Afferent Activity

### COMPARISON — UROTHELIUM-BASED VS MYOGENIC HYPOTHESES

|                     | Urothelium-based                                                                                                                     | Myogenic                                                                                                                                                        |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Origin              | Urothelium/suburothelium responds to mechanical, osmotic, inflammatory, chemical stimuli                                             | Detrusor smooth muscle                                                                                                                                          |
| Mechanism           | Altered receptors/channels + <b>mediator release onto adjacent afferents</b> → excess afferent traffic → urgency, micturition reflex | ↑ <b>spontaneous excitation + enhanced propagation</b> across too much of the bladder wall                                                                      |
| Supporting features | Intrinsic homeostatic aberration, unknown local trigger                                                                              | <b>Patchy denervation</b> (denervated cells upregulate receptors, altered membrane potential); structural change aiding spread; <b>exaggerated micromotions</b> |
| Link to urgency     | Augments fullness signal                                                                                                             | Micromotions distort wall → afferents sense filling without pressure change; <b>coordinated spread</b> → <b>urodynamic DO</b>                                   |

- **Other increased afferent signalling — pelvic organ cross-talk:** signals from other organs converge at **dorsal root ganglia, spinal cord or brain; bowel–bladder** best understood (shared thoracic and sacral innervation). Rectal distension or inflammation alters detrusor activity via **convergent C-fibres**; colonic stimulation worsens bladder pain in cystitis models. Clinically, **rectal distension changes capacity, sensation and DO; constipation and straining impair emptying and worsen storage and voiding symptoms.**
- **Abnormal handling of afferent signals:** continuous "afferent noise" reaches the central nervous system (CNS); normally most is filtered below consciousness. **If CNS modulation fails, normally unconscious sensations are perceived as urgency.**

## Other Pathophysiologic Mechanisms

- **Ischaemia-reperfusion** (atherosclerosis, arterial obstruction) → hypoxia, **oxidative stress** → damage to urothelium, muscle, microvasculature, nerves.
- **Chronic inflammation** — raised cytokines, chemokines, growth factors in tissue, urine and blood.
- **Urinary microbiome** differs in UUI/OAB — relevance to bladder-brain signalling unknown.

## Etiology and Attributable Risk Factors

- **Cause usually unknown and multifactorial.**
- **Aging** — clearly associated; multifactorial cellular and CNS change.
- **Sex** — higher prevalence (especially younger women) and **more UUI in women**.
- **BOO** — classic cause of DO in animal models, but in humans **not a direct cause: OAB prevalence similar in women (BOO rare) and men (BOO common); DO often persists after prostate surgery**. BOO may act indirectly, accelerating ageing changes.
- **Race/ethnicity** — differences exist; mechanisms unknown.
- **Psychological factors: depression and anxiety** most studied; embarrassment, shame, low self-esteem; **OAB wet carries more psychological burden than OAB dry**. Possible **serotonergic or hypothalamic-pituitary-adrenal axis** mechanisms; **stress** acts via this route; anxious women show **greater temporal summation of heat pain (central sensitisation marker)**.
- **Voiding behaviours: convenience ("just in case") voiding, delayed voiding, straining, positional preference** may cause OAB; **coping strategies (mapping toilets, altering plans, fluid restriction)** associated with stress and anxiety independent of symptoms and **perpetuate a negative bladder perception** even after objective improvement.
- **Wider associations:** bowel function, **metabolic syndrome**, hormonal status, psychosocial burden.

### KEY POINTS: PATHOPHYSIOLOGY AND ETIOLOGY

- Afferent information is integrated at multiple CNS levels; **main regulation at midbrain/brainstem**.
- **Failed high-level integration may let normally unconscious sensations be felt as urgency.**
- The two bladder-centric theories are **urothelium-based** and **myogenic**.
- **BOO and OAB are probably separate consequences of ageing rather than cause and effect.**

**Concept** — *urgency is the **perceived output of an afferent system** — too much signal from urothelium, muscle or neighbouring organs, or too little central filtering — which is why **DO, obstruction and bladder findings correlate only loosely with OAB** and why psychological state and behaviour modulate it.*

## CLINICAL ASSESSMENT

### Initial Evaluation

- OAB is a **symptom-based diagnosis**; **empiric treatment** may follow history and non-invasive testing.
- **Minimum (AUA/SUFU non-neurogenic OAB guideline): careful history, physical examination, urinalysis**; European Association of Urology (EAU) male and female LUTS guidelines agree and encourage **validated questionnaires and bladder diaries**.

## History

- **Each OAB symptom and its bother**; other storage (SUI) and **voiding symptoms suggesting BOO**; post-micturition symptoms, dysuria, haematuria. Questionnaires help but are limited by **health literacy and language**.
- **Fluid intake** — caffeine, polydipsia; patients may already restrict.
- **Occult neurologic disease** — new OAB with **erectile dysfunction or tremor**, spinal surgery/disc disease.
- Obstetric/gynaecologic, surgical, radiotherapy, **bowel** (faecal incontinence 7–15% in general population, ~50% in women with OAB) and medication history.
- **Comorbidities** that cause LUTS (polyuric diabetes), **relatively contraindicate antimuscarinics** (uncontrolled closed-angle glaucoma, cognitive impairment, retention, impaired gastric emptying) or  $\beta_3$  agonists (uncontrolled hypertension, arrhythmia), or  $\uparrow$  urine output (obstructive sleep apnoea, psychogenic polydipsia, lithium-induced diabetes insipidus).

## Physical Examination

- **Abdomen, pelvis, extremities, basic neurology, cognition and frailty** — even when the diagnosis seems straightforward.
- General appearance: gait, mobility, stature, frailty (performance status, neurology).
- Abdomen: scars, masses, hernias, tenderness, **suprapubic distension**.
- **Lower-limb oedema** — fluid shifts driving nocturia.
- **Pelvic/genital examination ( $\pm$  digital rectal examination in men)** — the most important component: women — pelvic floor tone, pain, **prolapse**, urethral prolapse/caruncle, labial adhesions, **vaginal atrophy (oestrogen status)**, supine stress test; men — **prostate size and abnormality**; all — **perineal sensation and rectal tone**.
- **Cognition** — observe during interview; **Mini-Mental State Examination** if concerned.
- **Frailty** predicts poor surgical outcome; **Timed Up and Go Test** (rise, walk 10 feet) is quick.

## Urinalysis

- **Mandatory** — exclude UTI, haematuria, pyuria; culture if infection suspected; **microhaematuria  $\rightarrow$  full evaluation**.

## Additional Assessment

### Other Laboratory Testing

- **Renal function not routine** (review if available); **prostate-specific antigen (PSA)** in men if it alters management; **culture** for acute irritative symptoms with a negative dipstick (dipstick misses low counts).

### Postvoid Residual Volume

- **Not required before lifestyle, behavioural or oral therapy in uncomplicated patients.**
- **Measure PVR in complicated patients:** retention risk, obstructive/voiding symptoms (**especially men**), prior anti-incontinence or prostate surgery, neurologic disease.
- **Portable bladder ultrasound preferred**; catheterisation if unavailable.

### Bladder Diary

- Records **intake and voiding**: useful when history is poor, recall unreliable, or **polyuria** suspected; guides behavioural change; **objective response measure**; self-monitoring itself therapeutic.
- **3–7 days ideal; at least 24–48 h** (burden); time, volume, leakage, circumstances, urgency grade.

- **OAB diary = frequent small-volume voids.** Daytime frequency counts voids while awake, including the last before sleep and the first on rising. Maximum voided volume = functional capacity (tracks OAB and DO severity). Electronic/smartphone diaries improve compliance.

**TABLE 110.4 BLADDER SENSATION SCALE USED IN THE INTERNATIONAL CONSULTATION ON INCONTINENCE QUESTIONNAIRE BLADDER DIARY (BRIGHT ET AL., 2012)**

| Grade | Meaning                                                          |
|-------|------------------------------------------------------------------|
| 0     | No sensation — voided for <b>social reasons</b> ("just in case") |
| 1     | <b>Normal desire, no urgency</b>                                 |
| 2     | Urgency that <b>passed before reaching the toilet</b>            |
| 3     | Urgency, reached toilet still urgent, <b>no leakage</b>          |

- **Trap:** grade 0 is **convenience voiding**, not a mild symptom; leakage is recorded separately, not as a sensation grade.

### Patient-Reported Outcome Measures and Symptom Questionnaires

- PROMs capture frequency, severity and bother — for diagnosis and **monitoring response**.
- **International Consultation on Incontinence (ICI)** grades PROMs **A–C** (+ for published content validity); **ICIQ modules** (from 1996) — standard, modular; **pair a symptom module with its HRQoL module**.
- For OAB: **ICIQ-MLUTS / ICIQ-FLUTS (sex-specific general)**, **ICIQ-OAB + ICIQ-OABqol**, core **ICIQ-UI**.
- **AUA/SUFU recommends: Urogenital Distress Inventory (UDI)**, **UDI-6**, **Incontinence Impact Questionnaire (IIQ)**, **OAB-q**.
- **LURN Symptom Index** (Lower Urinary Tract Research Network) — **both sexes, full LUTS spectrum**; 29-item (27 + 2 sex-specific) and 10-item forms.

**TABLE 110.3 DESCRIPTION OF COMMON PATIENT-REPORTED OUTCOME MEASURES FOR OVERACTIVE BLADDER, URGE INCONTINENCE, AND LOWER URINARY TRACT SYMPTOMS**

| Group               | Instrument (items)                                               | Measures                                                                                 |
|---------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>OAB-specific</b> | <b>ICIQ-OAB (4); OAB-SS</b> Overactive Bladder Symptom Score (7) | Frequency, nocturia, urgency, UUI                                                        |
|                     | <b>ICIQ-OABqol (26); OAB-q short form (19)</b>                   | HRQoL in OAB (coping, worry, sleep, social)                                              |
|                     | <b>OAB-V8 (8)</b>                                                | <b>Primary-care screening</b>                                                            |
| <b>General LUTS</b> | <b>ICIQ-FLUTS (12; long 18); ICIQ-MLUTS (13; long 23)</b>        | <b>Sex-specific</b> LUTS and bother                                                      |
|                     | <b>LURN SI-29 (27 + 2); LURN SI-10 (10)</b>                      | Both sexes: urgency, UI, voiding difficulty, nocturia, pain, frequency, post-micturition |
|                     | <b>LUTSS (14); LUSQ</b> Leicester (10); <b>B-SAQ (8, women)</b>  | LUTS severity ± bother                                                                   |
|                     | <b>ICIQ-LUTSqol (22); Leicester Impact Scale (21)</b>            | LUTS-related QoL                                                                         |
| <b>Incontinence</b> | <b>ICIQ-UI short form (4)</b>                                    | Frequency, amount, perceived cause                                                       |
|                     | <b>I-QOL (22); IIQ-7 (7); UISS (10)</b>                          | UI impact on QoL                                                                         |
|                     | <b>UDI-6 (6)</b>                                                 | LUTS and UI in women                                                                     |
|                     | <b>ISS (8); PRAFAB (5)</b>                                       | Severity; treatment effect in women                                                      |
|                     | <b>QUID (6)</b>                                                  | <b>Distinguishes type of UI</b> in women                                                 |

- **Pattern:** instruments separate **symptoms** from **HRQoL impact** — pair one of each; most incontinence tools are **female-validated**; only LURN SI, LUTSS and the Leicester tools span both sexes.

## Advanced Evaluation

- **Urodynamics, cystoscopy and renal/bladder ultrasound are not for the initial work-up of the uncomplicated patient.**
- **Urodynamics indicated when compliance, DO or other findings will decide for or against an invasive, irreversible or morbid treatment, or after failure of multiple treatments; also suspected BOO (e.g. OAB after an outlet procedure).** May improve satisfaction but **not proven to improve outcome for the index patient** (woman, low PVR, no significant prolapse, no neurologic disease, minimal SUI).
- **Aim: reproduce the symptoms;** two OAB diagnoses — **DO** and **early/exaggerated filling sensation** ("early and persistent desire to void" when DO absent).
- **Stopping antimuscarinics before testing:** stop → best chance of seeing DO; continue → explains refractory symptoms.
- **DO vs symptoms (Fig. 110.5):** DO correlates **best with UII, worst with nocturia**. OAB dry: DO in ~60–70% of men and 40–50% of women; OAB wet: DO in nearly all men, but absent in 30–40% of women with reported UII (SUI and pre-emptive voiding misread as urgency).
- **Technique:** use provocations (running water, standing); **fast filling provokes DO** and may obscure other parameters; test **seated/standing** (symptoms occur upright); if severe low-volume DO hides SUI → **second fill supine** (reduces DO).
- **Always assess BOO and emptying:** (1) OAB treatment may precipitate retention — **each 10 mL of PVR ↑ risk of catheterisation after onabotulinumtoxinA by ~27%**; (2) severe BOO associated with DO — OAB may not improve unless BOO treated, yet **relief of BOO does not guarantee improvement (DO in up to ~60% after TURP)**.

### KEY POINTS: OVERACTIVE BLADDER CLINICAL ASSESSMENT

- **Minimum: history, examination, urinalysis.**
- Consider **malignancy, neurologic disease, systemic disease, elevated PVR.**
- Urine culture and PVR at the provider's discretion when indicated.
- **Diaries and validated PROMs** clarify symptoms and track response.
- **No urodynamics, cystoscopy or ultrasound in uncomplicated initial work-up.**
- **Urodynamics when findings will change an invasive/irreversible decision, or after multiple failed treatments.**

**Concept** — *because OAB is a **symptomatic diagnosis of exclusion**, assessment is built to **exclude what mimics it** (infection, malignancy, neurology, retention, polyuria) with minimal testing; invasive testing is justified **only when its answer will change an irreversible decision**.*

## MANAGEMENT

- **OAB is chronic, affects QoL, and is not life-threatening** — counsel in that context. **Choosing no treatment is acceptable** after counselling (benefits may not outweigh costs; severe cognitive impairment).
- Address **mobility limits** (correcting leakage will not correct immobility); **incontinence-associated dermatitis** (pain, infection, pressure ulcers) may justify more aggressive treatment.
- **Set goals and realistic expectations** — improvement, not cure; gradual and irregular; goals may change; **critical before invasive therapy**.
- **Education** empowers and itself improves perception; **substantial placebo response in OAB trials** shows the power of non-pharmacologic intervention.



- **AUA/SUFU lines (first–fourth) balance benefit against invasiveness, adverse events and reversibility — not a mandatory linear path;** lines are often combined.

## Conservative Management

- Behavioural therapy and lifestyle: **bladder training, urge-control strategies, pelvic floor muscle therapy (PFMT), weight loss, fluid management** — mostly studied for UI in women; improvement, rarely resolution.
- **Fluid: ~25% reduction meaningfully improves symptoms** (cost: thirst, underhydration); many already restrict.
- **Caffeine reduction improves urgency and frequency but not consistently UI** — let the patient judge the trade-off.
- **Weight loss:** reduces SUI and UUI (intensive programme in obese women); **bariatric surgery ~7–8% lower UI risk per 5% weight lost.**
- **PFMT and bladder training: incontinence episodes fall 50–80%, negligible risk; combine with drugs** according to preference.

## Pharmacologic Management

### COMPARISON — ORAL DRUG CLASSES FOR OAB

|                            | Antimuscarinics                                                                                | β3 agonists (e.g. mirabegron)                | α1-blockers (men)                       |
|----------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------|
| Role                       | Established; cure ↑ 20–70% over placebo; <b>high withdrawal</b>                                | <b>Preferred agents (SUFU)</b>               | <b>First-line for male voiding LUTS</b> |
| Key adverse effects        | Dry mouth, constipation, <b>cognitive impairment</b> , blurred vision                          | —                                            | —                                       |
| Relative contraindications | Uncontrolled closed-angle glaucoma, cognitive impairment, retention, impaired gastric emptying | <b>Uncontrolled hypertension, arrhythmia</b> | —                                       |
| Formulation                | <b>Extended-release preferred</b> (less dry mouth and CNS effect)                              | —                                            | Add antimuscarinic/β3 for storage LUTS  |
| Dementia                   | <b>Chronic use (&gt;3 months) → 10–65% ↑ new-onset dementia</b>                                | Avoids anticholinergic burden                | —                                       |

- **SUFU white paper: prefer β3 agonists, or extended-release trospium, darifenacin or fesoterodine** (limited CNS penetration); **shared decision-making** before anticholinergics.
- **Responses are idiosyncratic** — titrate; if inadequate or intolerable, change dose or agent.
- **Men with BPH + OAB:** α1-blocker + antimuscarinic beats either alone, but **acute retention up to ~3% → caution when PVR is raised.**
- **Initial failure → dose change, different agent or combination** before invasive work-up.

## Procedural or Surgical Management

- **"Third-line" (AUA/SUFU): after failure of or unsuitability for conservative and drug therapy** — may be chosen earlier for medical, social or personal barriers; more invasive, more adverse events.
- **Frail elderly:** respond well, but **frailty ↑ morbidity** (functional decline, discharge to care, high 1-year morbidity even after minor procedures).

**COMPARISON — THIRD-LINE THERAPIES FOR REFRACTORY OAB**

|               | PTNS (percutaneous tibial nerve stimulation)                        | Intradetrusor onabotulinumtoxinA                                               | SNS (sacral nerve stimulation)                           |
|---------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------|
| Efficacy      | Response ~60–80%                                                    | UI episodes ↓ ~59%, urgency ↓ ~38%, frequency ↓ ~29%; ~¼ completely dry        | 5-year success ~70–80% (↓ UUI, ↓ voids)                  |
| Main drawback | Weekly office sessions for 12 weeks; response delayed to weeks 8–10 | UTI ~20%; CIC 0–12% in trials, retention ~20% women, up to 35% men in practice | Revision >30% at 5 years; battery replacement inevitable |
| Advantage     | Minimal adverse events                                              | Most cost-effective (100 U)                                                    | Rechargeable, longer battery, MRI-compatible devices     |
| Cost          | —                                                                   | Lowest of third-line                                                           | Costliest over 10 years                                  |

- Choice (including declining treatment) is by patient preference after informed discussion of benefits and drawbacks.

**Augmentation Cystoplasty or Urinary Diversion**

- Rarely for intractable OAB; evidence mainly neurogenic → little known of QoL in non-neurogenic OAB; weigh against containment (pads) or indwelling catheter.

**Approach**

- ICI (sixth consultation) grades evidence and gives care pathways for men, women and the frail elderly (Fig. 110.6); AUA/SUFU algorithm and SUFU OAB care pathway concur.
- Common sequence: assess to confirm OAB and exclude alternatives → educate → conservative (lifestyle, behavioural, PFMT) → pharmacologic → invasive (tibial stimulation, SNM, chemodenervation); reassess symptoms, side effects and preferences at every step.
- Minimum trial before judging efficacy: 4–12 weeks for behavioural and for drug therapy.

**FIG. 110.6 ICI ASSESSMENT AND MANAGEMENT — MEN, WOMEN AND FRAIL ELDERLY**

|                    | Men                                                                                                                                                            | Women                                                                                                                                                   | Frail elderly                                                                                                                                                                                                                        |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presentation       | Urgency/frequency ± incontinence                                                                                                                               | Incontinence with mixed symptoms                                                                                                                        | Active case finding                                                                                                                                                                                                                  |
| Initial assessment | Symptom score (incl. frequency–volume chart), QoL, urinalysis ± culture; <b>abdominal, rectal, sacral, neurologic examination</b> ; pelvic floor function; PVR | Same general assessment; <b>abdominal, pelvic, perineal examination</b> ; cough test; <b>oestrogen status</b> ; voluntary pelvic floor contraction; PVR | <b>Treat reversible comorbidities</b> ; goals incl. <b>carer preference</b> ; <b>targeted examination incl. cognition</b> , neurologic and rectal; urinalysis; <b>frequency–volume chart or wet checks, especially with nocturia</b> |
| Initial management | Lifestyle; PFMT ± biofeedback; bladder training; products; <b>antimuscarinic (OAB ± UUI) and α-blocker if also BOO</b>                                         | Lifestyle; PFMT for SUI or OAB; bladder retraining; <b>antimuscarinic</b>                                                                               | Lifestyle, behavioural therapy; <b>trial of antimuscarinic</b> ; <b>treat significant PVR</b>                                                                                                                                        |

|            | Men                                                                                                                                                                                                                                                                 | Women                                                                                                                                                                                                                | Frail elderly                                                                          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| On failure | <b>Specialist:</b> urodynamics, urinary tract imaging, urethrocystoscopy if indicated; <b>treat major component first</b> → $\alpha$ -blocker/5 $\alpha$ -reductase inhibitor or correct BOO; antimuscarinic; <b>neuromodulation</b> ; intermittent catheterisation | <b>Specialist:</b> assess prolapse, imaging, urodynamics; <b>treat most bothersome symptom first</b> → SUI surgery (bulking, tapes/slings, colposuspension) <b>or</b> botulinum toxin, neuromodulation, augmentation | Reassess comorbidity and function; <b>refer per patient preference and comorbidity</b> |

- **Pattern:** men — **identify and treat obstruction**; women — **stress vs urgency, treat the most bothersome first**; frail elderly — **comorbidity, function and carer goals before escalation**.

#### KEY POINTS: CLINICAL MANAGEMENT

- **Conservative:** caffeine and fluid reduction, weight loss, bladder training, PFMT.
- **Pharmacologic:** antimuscarinics (oral or transdermal) or  $\beta_3$  agonists — **caution with chronic antimuscarinics (dementia risk)**.
- **Procedural:** tibial neuromodulation, botulinum toxin, sacral neuromodulation — **shared decision-making**.
- **Augmentation or diversion** for the refractory, weighed against containment or indwelling catheter.

**Concept** — OAB treatment escalates by **invasiveness and reversibility, not by severity alone**, because the condition is benign and chronic; every step is gated by **the patient's goals and tolerance**, and **no treatment remains a legitimate endpoint**.